DBscreamZ

Pitch-tracking formant scream synthesiser

User manual
& preset field guide

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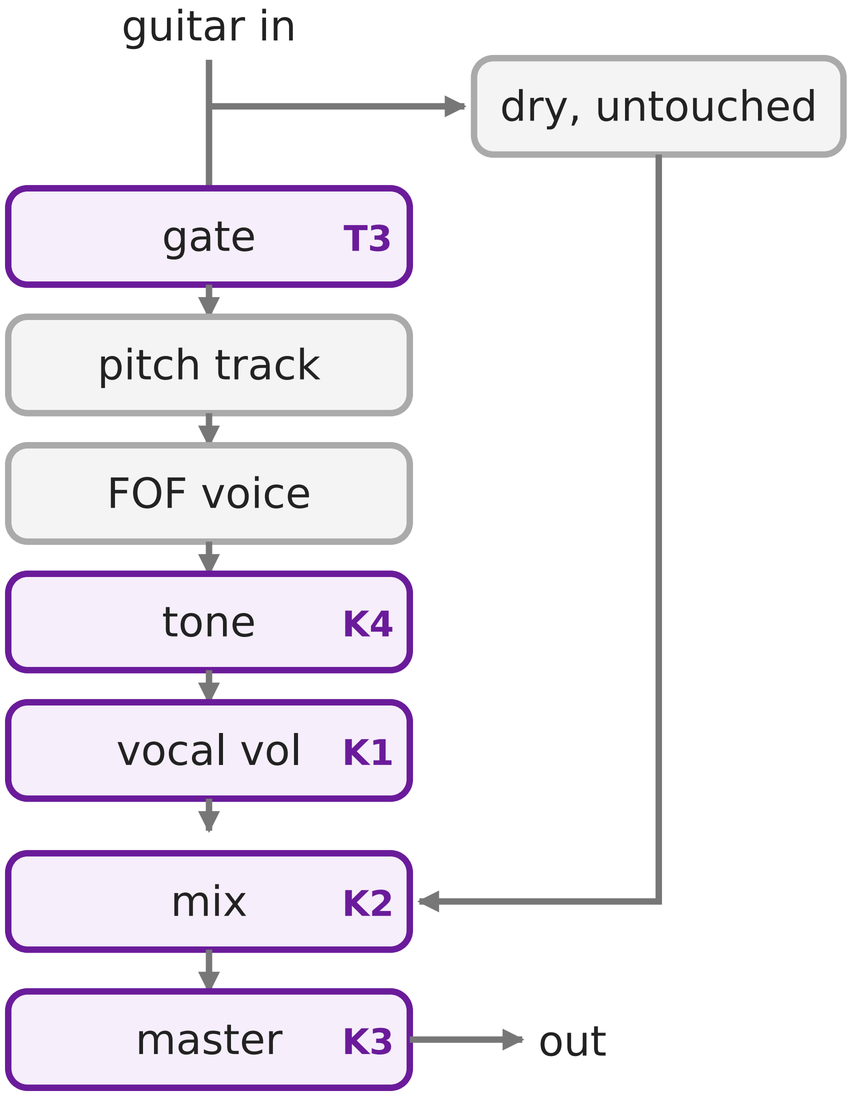
1 What this is

Play a note. A synthesised voice screams it back at you, tracking your pitch as you bend, slide and vibrato.

The voice is not a sample and not a filter draped over your guitar. It is built from scratch every block: a pitch tracker follows your fundamental, and a bank of formant grains rebuilds a vowel at that pitch. Three formants (F1, F2, F3) decide which vowel and how bright it is. There is no noise anywhere in the voice path, which is why it screams rather than hisses.

Your dry signal is never processed. It is passed through untouched and crossfaded against the voice at the end, so Mix at 7:00 is your guitar, exactly as it went in.

Signal flow



The gate watches your input envelope and decides when the voice may sound. It never gates your dry signal.

2 First sound

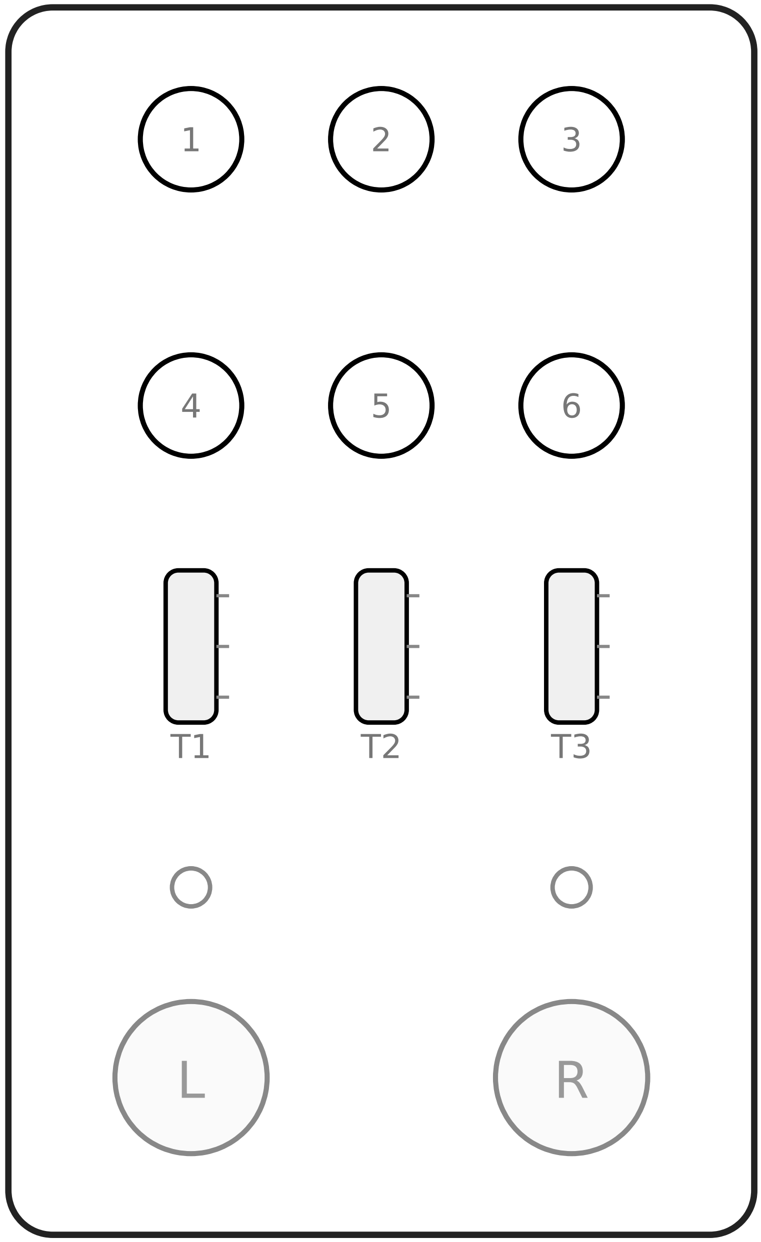
1. Guitar into **IN**, amp into **OUT**, 9 V DC into the barrel jack. The pedal boots **bypassed**, both LEDs off.
2. Middle toggle **up** (Set 1). Put the other two toggles in the **middle**.
3. Tap the **RIGHT** footswitch. The right LED goes solid: Wukong is engaged.
4. Play single notes, cleanly, one at a time. The voice follows your pitch.
5. Nothing screaming? Turn **knob 2 (Mix)** clockwise and set the **right toggle to the middle**.
6. Now stomp **both** footswitches at once and hold. That is Charge mode.

> **ONE NOTE AT A TIME**
>
> Pitch tracking needs a single fundamental. Chords, open strings ringing under a lead line, and heavy pick noise all confuse it. Mute what you are not playing.

Tap the same footswitch again to bypass. Tap the **LEFT** footswitch instead for the other voice on this page (Prince). Flip the middle toggle **down** for Set 2: Rice on the right, Piccolo on the left.

Everything you change is live immediately and lost at power off until you save it. Saving is in section 5.

3 The controls



Six knobs in two rows of three, numbered left to right: **1 2 3** on top, **4 5 6** below. Three toggles, each up, middle or down. Two footswitches, an LED [light-emitting diode] above each.

Knob travel

Knobs run **7:00** fully anticlockwise, through **12:00** at centre, to **5:00** fully clockwise. Every position in this booklet is a clock face.

What the knobs do

	Menu 1	Menu 2	Menu 3
K1	Vocal vol	F1	F3
K2	Mix	F1 bandwidth	F3 bandwidth
K3	Master vol	F1 amount	F3 amount
K4	Tone	F2	Voices
K5	Glide	F2 bandwidth	Detune
K6	Vocal size	F2 amount	Grain

Menu 3: **K4** steps through the eight voice counts in eight equal bands, so it always lands on a whole number. **K6** has a detent at **12:00** that reads as exactly 20 ms, the grain length every character ships with, so centring it always returns you to the factory texture.

What the menu 1 knobs do

Vocal vol (K1) is how loud the synthesised voice is on its own. **Mix** (K2) crossfades it against your untouched dry signal: 7:00 is the guitar alone, 5:00 is the voice alone. **Master** (K3) is the level of the pair leaving the pedal. All three are unity at 12:00.

Tone (K4) tilts the *voice* dark below 12:00 and bright above it, and never touches the dry. It has a detent at 12:00 that is exactly flat.

Glide (K5) is how long the voice takes to reach each new note, 0 to 300 ms. The taper is steep: the first two thirds of the travel covers 0 to 100 ms, so the whole usable range of slurs sits below 2:00.

Vocal size (K6) scales all three formants together, which is acoustically vocal tract length: the same character shouting the same vowel out of a physically bigger body. At **7:00** it is the character exactly as written, which is what every card in section 8 assumes. At **5:00** every formant is halved, the deepest it goes.

What the toggles do

	Up	Middle	Down
T1 Octave	+1	0	−1
T2 Page	Set 1	Freeform	Set 2
T3 Gate	high	medium	low

That is with no menu latched. While a menu *is* latched the toggles configure Charge mode instead, and nothing else. See section 6.

The gate decides how loud you must play before the voice speaks. High needs a firm attack and stays quiet between notes; low lets quiet playing through and hangs on longer.

4 The three menus

Six knobs, eighteen parameters. What a knob does depends on which menu is latched.

Menu	How to get there
1	Where you are with nothing latched
2	Hold the **RIGHT** footswitch. About 1 second in it latches under your foot: right LED blinks, left goes dark. Letting go changes nothing.
3	Hold the **LEFT** footswitch the same way. Left LED blinks, right goes dark.

A menu **latches**: it stays until you leave it.

- Tap the **other** footswitch to jump straight to the other menu.
- Tap the **blinking side's own** footswitch to drop back to menu 1.

> **KNOB PICKUP: NOTHING JUMPS**
>
> Every time you change menu, or recall a voice, all six knobs go **inert**. A knob does nothing until you **move it**, about a fingernail's nudge. The moment it moves it takes over its parameter, from wherever it is physically sitting.
>
> A knob you do not touch cannot hurt the sound. A knob you do touch grabs control at once: it does not wait for you to sweep back to the stored value.

What the formant controls mean

F1 and F2 decide which vowel the voice is shouting. Move them and the scream changes from an *ah* to an *ee* to an *oh*.

F3 mostly decides how bright and nasal it sounds.

Bandwidth is how wide each resonance is: narrow is more vocal, wide is more washed out. **Amount** is how much of that formant you get.

Voices is how many copies of the scream sing at once, one to eight, and **Detune** is how far apart they are tuned, in cents. Together they are the thickness of the voice: one voice is bare and focused, eight spread wide is a crowd of one person.

Grain is how long each grain of the voice lasts, from 4 to 40 ms. It sets texture rather than pitch: short grains are buzzy and rough, long ones smooth and vocal. There is no noise generator anywhere in this pedal, so every one of these controls shapes the voice itself.

5 Memory

The middle toggle picks a **page**. Each page has two **slots**, one per footswitch. Four stored voices.

T2	Page	Left	Right
Up	Set 1	Prince	Wukong
Mid	Freeform	no slots: the footswitches engage and bypass whatever the knobs are set to right now	
Down	Set 2	Piccolo	Rice

Those four are what it ships with. Save over any of them: the recipes to dial them back are in section 8.

Saving

Everything you tweak is live but volatile. To save:

1. Hold **one** footswitch past 1 second.
2. **While still holding it**, press the other.

The **held** side picks the slot. Hold RIGHT and press LEFT and you have saved to the **right** slot of the current page. Both LEDs blink three times.

> **FREEFORM CANNOT SAVE**
>
> It has no slot, so the save is refused and both LEDs give one short double-flicker. Pick Set 1 or Set 2 first.

Leave any latched menu before saving. Menus never save, and your edits survive the exit.

The held footswitch latches its own menu on the way past 1 second, the same as any hold. The second press drops that menu again and saves, so you finish where you began.

6 Charge mode

With a voice engaged and **no menu latched**, stomp **both** footswitches together and hold. The scream charges: gain swells, pitch sweeps, the voice grows, and the LEDs alternate faster and faster as it builds. Release and it winds down.

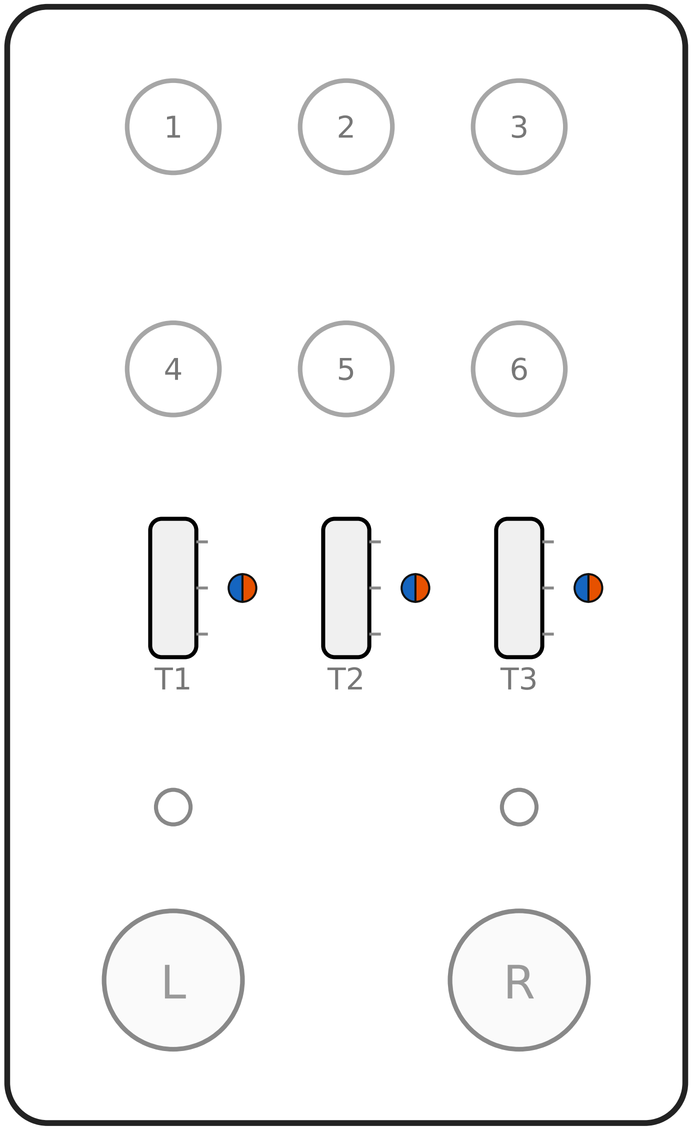
At a **full charge every row you have switched on reaches the top of its range**: the gain is all the way up, the pitch has swept two octaves, and the voice has grown to its deepest. On the amount rows, gain and size, the middle position gets you halfway there from wherever the voice already sits and the up position takes it the whole way. Tone is a direction rather than an amount, so both of its positions go the whole way: darker is fully dark, brighter fully bright.

It lasts exactly as long as you hold it. No latch, no timeout.

Engaged is the whole condition. The same two-stomp hold while *bypassed* is the firmware-update gesture instead, so charge can never reach it and it can never reach charge. See section 11.

Charge is configured on the **toggles**, while a menu is latched. Latch menu 2 (hold RIGHT) for the orange settings, menu 3 (hold LEFT) for the blue ones. The knobs do nothing here.

Factory charge settings



Every toggle centred. Each dot is half orange and half blue because both menus want that switch in the middle: orange on the right, blue on the left.

This configuration is **global**: one setting shared by every voice, kept across power cycles, and never touched by saving or recalling. That is why it is not on the character cards.

With menu 2 latched

	Up	Middle	Down
T1 Gain	Above 9000!	on	off
T2 Charge time	Birit Spomb ~6 s	Hamekameka ~2.5 s	punch ~0.75 s
T3 Decay	fast	slow	off (instant)

With menu 3 latched

	Up	Middle	Down
T1 Pitch	rise 2 oct	fall 2 oct	off
T2 Tone	brighter	darker	off
T3 Size	full	half	off

Centred means the charge pitch *falls*. For the rising power-up, latch menu 3 and flick T1 up.

7 Reading a card

Each card is the pedal face with a pointer drawn out of every knob, one per menu, ending at the position that menu wants. **Twenty-one positions:** six knobs in each of three menus, plus three toggles. Set every one. None can be assumed: the knobs are wherever you left them.

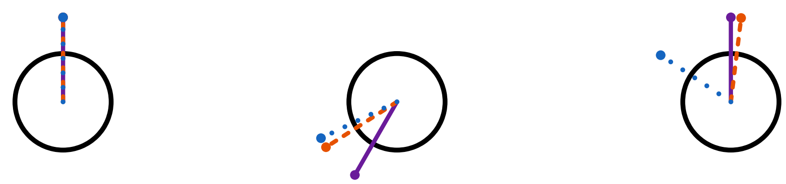
Menu 1, solid pointer
default layer

Menu 2, dashed pointer
hold RIGHT

Menu 3, dotted pointer
hold LEFT

Told apart by colour *and* by line, so a photocopy still works. Menus 2 and 3 wear the colour of the LED on the side you hold to reach them: orange for the right, blue for the left.

Every pointer sits at its exact position, including when menus agree. The line styles interleave, so where two or three menus want the same knob in the same place you get one pointer striped in their colours rather than three lines fighting over the same pixels.



All three agree. Knob 3 on every card: 12:00 in all three menus.

One apart, two close. Knob 2 on every card.

A tight cluster. Fling's knob 4, inside 14 minutes.

> **THE TABLE IS THE EXACT WORD**
>
> Pointers are drawn true, but a knob is a blunt instrument and two pointers a few minutes apart look like one. Each card is followed by the same twenty-one positions as a table. When the drawing is ambiguous, read the table.

Toggles on a card

Toggle dots sit to the right of the switch, level with the position they want.

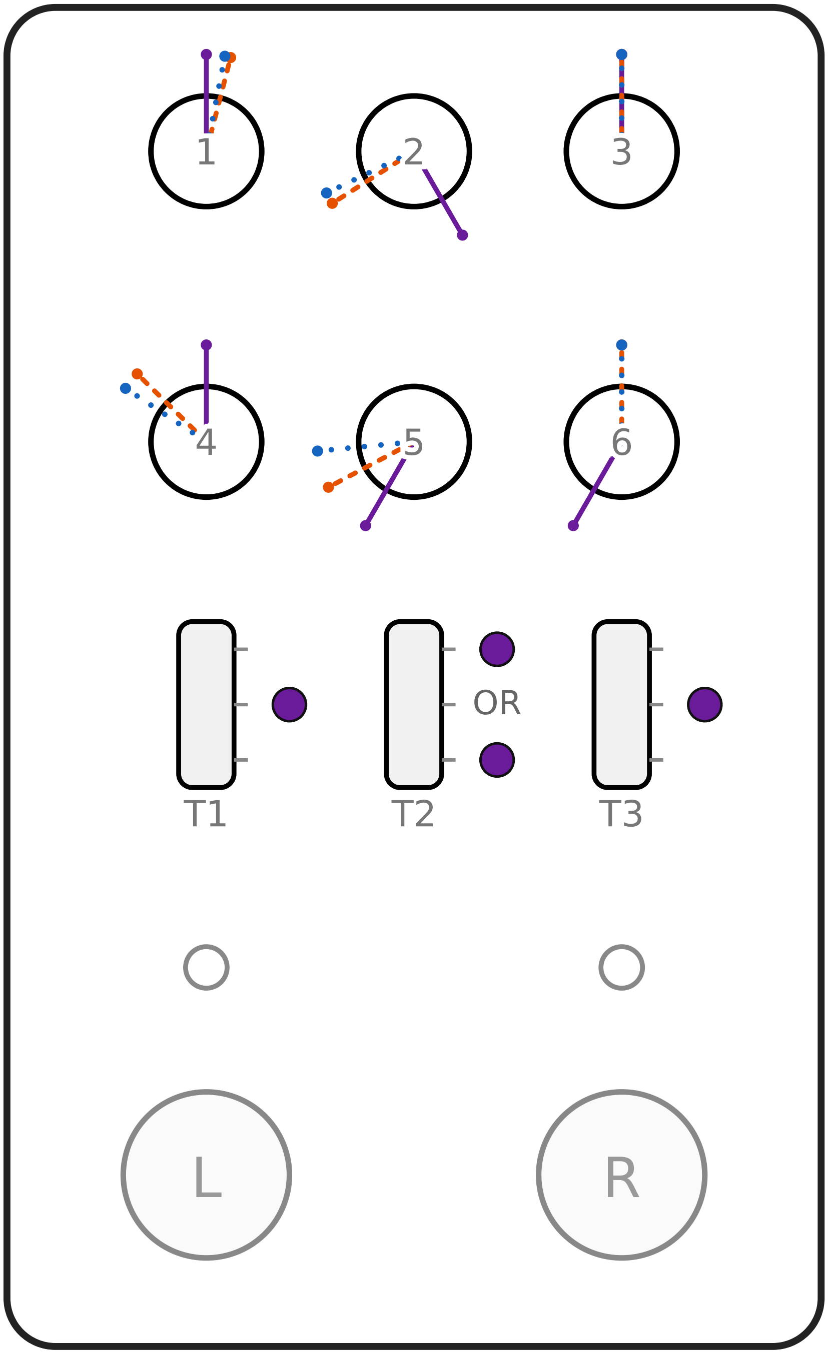
The middle toggle always shows **two** dots, marked OR. A voice can live on either Set, and which one you pick is up to you. It cannot live on Freeform, because Freeform has no slot to save into.

The other two toggles show one dot each. Every character wants octave 0 and the medium gate; both are the middle position.

Four of the eight have a factory home. The other four are recipes: dial them in and save them wherever you like.

Wukong *The default shout*

Factory: Set 1, RIGHT slot



Mid-placed formants over the reference three-voice stack at the default grain. The straightest read of the effect: loud, open, clean. Start here.

Wukong: every position

Menu 1 (default)

K1	Vocal vol	unity	50% · 12:00
K2	Mix	full voice	100% · 5:00
K3	Master vol	unity	50% · 12:00
K4	Tone	flat	50% · 12:00
K5	Glide	0 ms	0% · 7:00
K6	Vocal size	as written	0% · 7:00

Menu 2 (hold RIGHT)

K1	F1	858.4 Hz	55% · 12:29
K2	F1 bandwidth	32.5 Hz	9% · 7:55
K3	F1 amount	1.0	50% · 12:00
K4	F2	1234 Hz	35% · 10:29
K5	F2 bandwidth	47.5 Hz	11% · 8:04
K6	F2 amount	1.0	50% · 12:00

Menu 3 (hold LEFT)

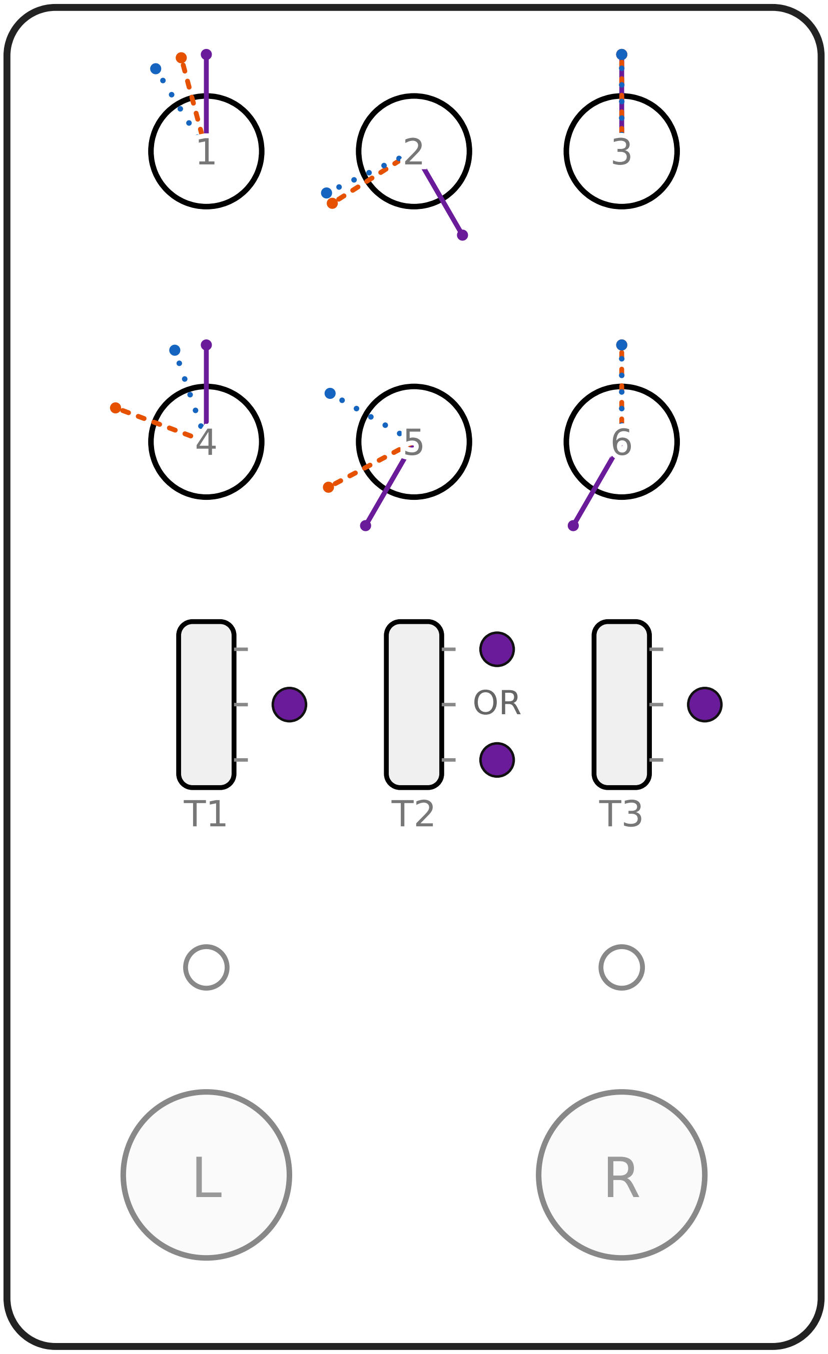
K1	F3	3111.7 Hz	54% · 12:22
K2	F3 bandwidth	62.5 Hz	12% · 8:09
K3	F3 amount	1.0	50% · 12:00
K4	Voices	3	31% · 10:07
K5	Detune	11 ct	18% · 8:49
K6	Grain	20 ms	50% · 12:00

Toggles

T1	Octave	0	Middle
T2	Memory page	Set 1 or Set 2	Up or Down
T3	Gate	medium	Middle

Prince *Clenched teeth*

Factory: Set 1, LEFT slot



Formants well below Wukong under a wider four-voice stack, so it lands darker and rougher. Reads as effort rather than power.

Prince: every position

Menu 1 (default)

K1	Vocal vol	unity	50% · 12:00
K2	Mix	full voice	100% · 5:00
K3	Master vol	unity	50% · 12:00
K4	Tone	flat	50% · 12:00
K5	Glide	0 ms	0% · 7:00
K6	Vocal size	as written	0% · 7:00

Menu 2 (hold RIGHT)

K1	F1	741.6 Hz	45% · 11:30
K2	F1 bandwidth	32.5 Hz	9% · 7:55
K3	F1 amount	1.0	50% · 12:00
K4	F2	1066 Hz	27% · 9:41
K5	F2 bandwidth	47.5 Hz	11% · 8:04
K6	F2 amount	1.0	50% · 12:00

Menu 3 (hold LEFT)

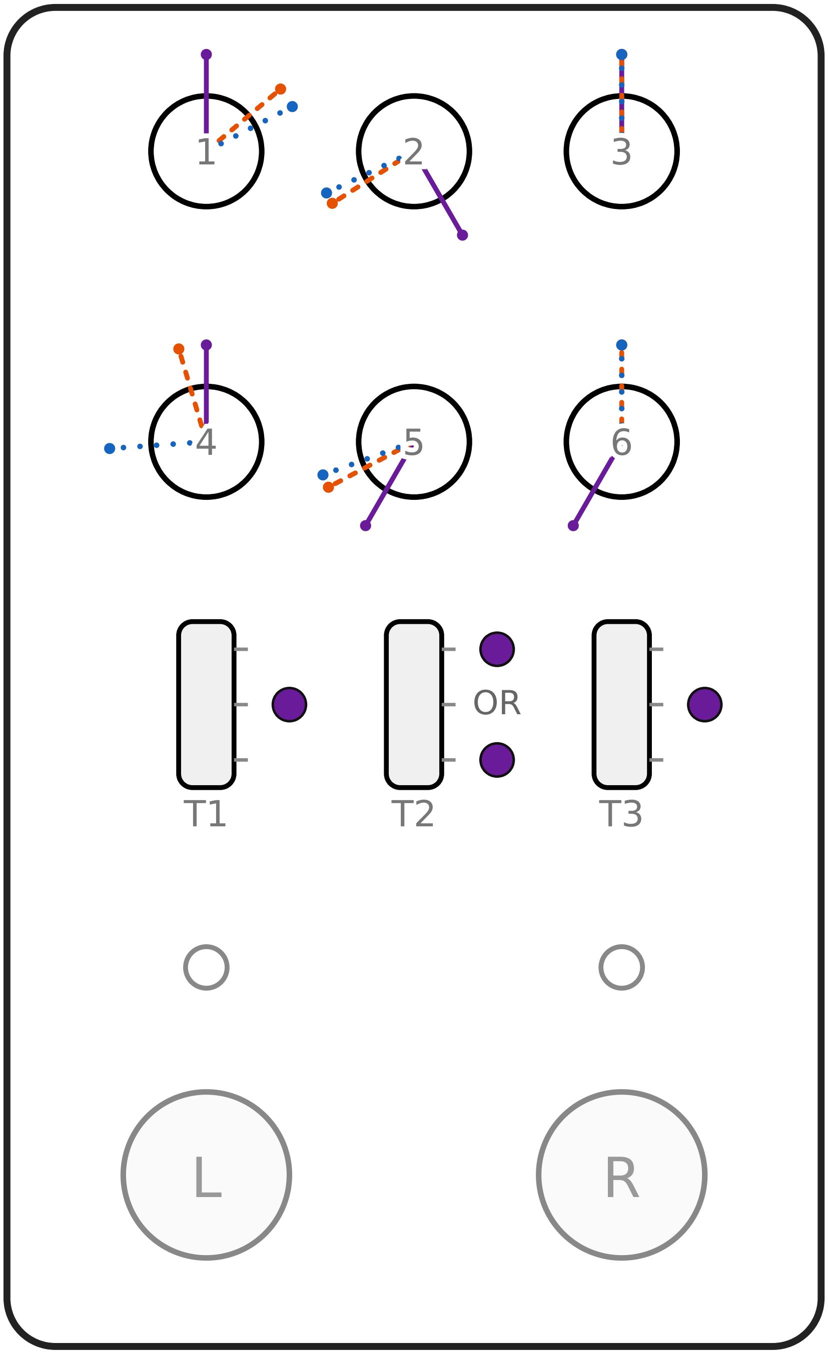
K1	F3	2688.3 Hz	40% · 10:57
K2	F3 bandwidth	62.5 Hz	12% · 8:09
K3	F3 amount	1.0	50% · 12:00
K4	Voices	4	44% · 11:22
K5	Detune	18 ct	30% · 10:00
K6	Grain	20 ms	50% · 12:00

Toggles

T1	Octave	0	Middle
T2	Memory page	Set 1 or Set 2	Up or Down
T3	Gate	medium	Middle

Rice *Clean and glassy*

Factory: Set 2, RIGHT slot



The highest F3 of the eight over the thinnest stack in the factory set: two voices, barely detuned. Bright and hard-edged. Cuts without sounding strained.

Rice: every position

Menu 1 (default)

K1	Vocal vol	unity	50% · 12:00
K2	Mix	full voice	100% · 5:00
K3	Master vol	unity	50% · 12:00
K4	Tone	flat	50% · 12:00
K5	Glide	0 ms	0% · 7:00
K6	Vocal size	as written	0% · 7:00

Menu 2 (hold RIGHT)

K1	F1	1000 Hz	67% · 1:40
K2	F1 bandwidth	32.5 Hz	9% · 7:55
K3	F1 amount	1.0	50% · 12:00
K4	F2	1437.5 Hz	45% · 11:27
K5	F2 bandwidth	47.5 Hz	11% · 8:04
K6	F2 amount	1.0	50% · 12:00

Menu 3 (hold LEFT)

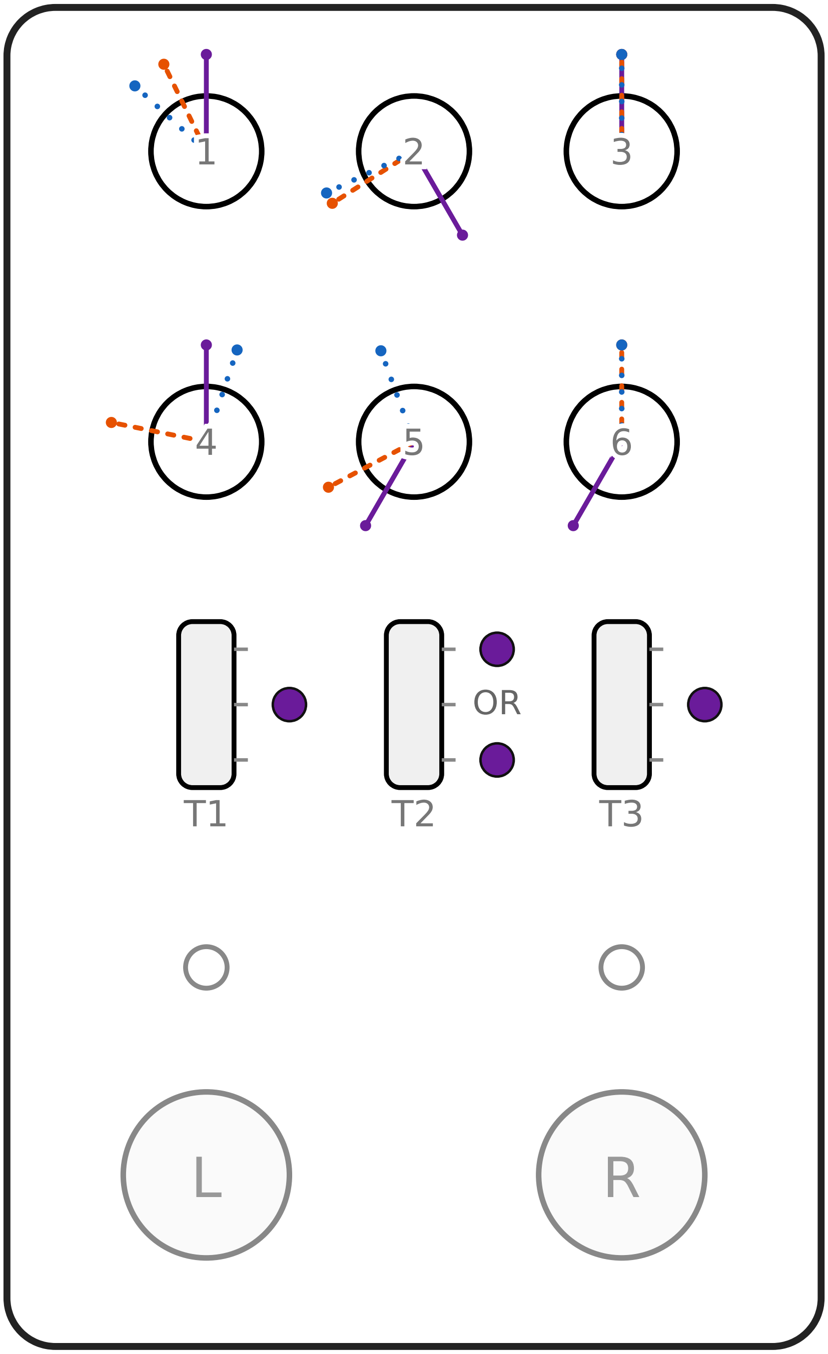
K1	F3	3625 Hz	71% · 2:05
K2	F3 bandwidth	62.5 Hz	12% · 8:09
K3	F3 amount	1.0	50% · 12:00
K4	Voices	2	19% · 8:52
K5	Detune	8 ct	13% · 8:20
K6	Grain	20 ms	50% · 12:00

Toggles

T1	Octave	0	Middle
T2	Memory page	Set 1 or Set 2	Up or Down
T3	Gate	medium	Middle

Piccolo *Low growl*

Factory: Set 2, LEFT slot



The lowest formants in the factory set under the thickest factory stack, five voices spread wide. Thick and throaty rather than piercing.

Piccolo: every position

Menu 1 (default)

K1	Vocal vol	unity	50% · 12:00
K2	Mix	full voice	100% · 5:00
K3	Master vol	unity	50% · 12:00
K4	Tone	flat	50% · 12:00
K5	Glide	0 ms	0% · 7:00
K6	Vocal size	as written	0% · 7:00

Menu 2 (hold RIGHT)

K1	F1	697.6 Hz	41% · 11:08
K2	F1 bandwidth	32.5 Hz	9% · 7:55
K3	F1 amount	1.0	50% · 12:00
K4	F2	1002.8 Hz	24% · 9:23
K5	F2 bandwidth	47.5 Hz	11% · 8:04
K6	F2 amount	1.0	50% · 12:00

Menu 3 (hold LEFT)

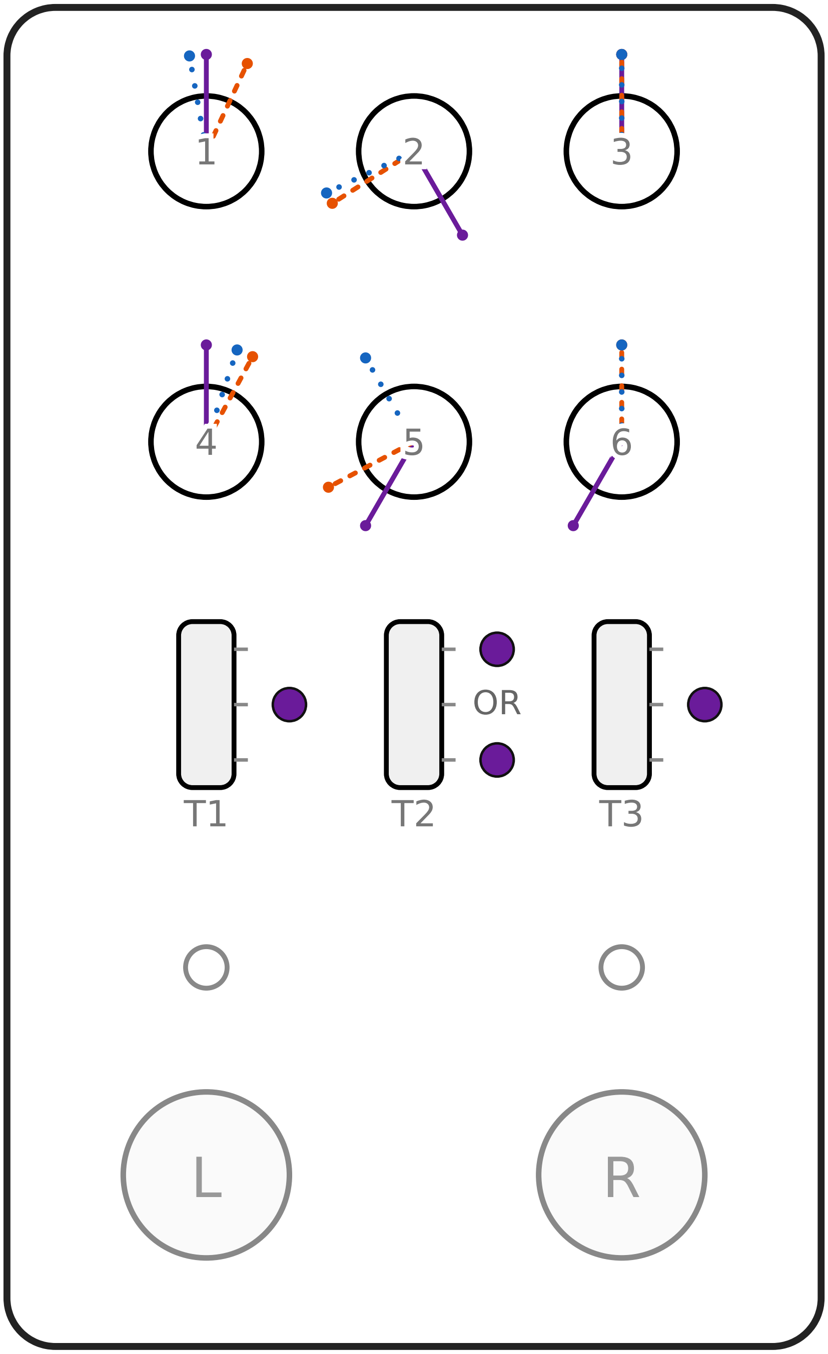
K1	F3	2528.8 Hz	34% · 10:25
K2	F3 bandwidth	62.5 Hz	12% · 8:09
K3	F3 amount	1.0	50% · 12:00
K4	Voices	5	56% · 12:37
K5	Detune	26 ct	43% · 11:20
K6	Grain	20 ms	50% · 12:00

Toggles

T1	Octave	0	Middle
T2	Memory page	Set 1 or Set 2	Up or Down
T3	Gate	medium	Middle

Boo *Rubbery and hollow*

No factory slot: save it to any page



A wide F1-to-F2 gap over a low F3, thickened by five widely detuned voices. A big soft body rather than an edge.

Boo: every position

Menu 1 (default)

K1	Vocal vol	unity	50% · 12:00
K2	Mix	full voice	100% · 5:00
K3	Master vol	unity	50% · 12:00
K4	Tone	flat	50% · 12:00
K5	Glide	0 ms	0% · 7:00
K6	Vocal size	as written	0% · 7:00

Menu 2 (hold RIGHT)

K1	F1	900 Hz	58% · 12:50
K2	F1 bandwidth	32.5 Hz	9% · 7:55
K3	F1 amount	1.0	50% · 12:00
K4	F2	1750 Hz	60% · 12:57
K5	F2 bandwidth	47.5 Hz	11% · 8:04
K6	F2 amount	1.0	50% · 12:00

Menu 3 (hold LEFT)

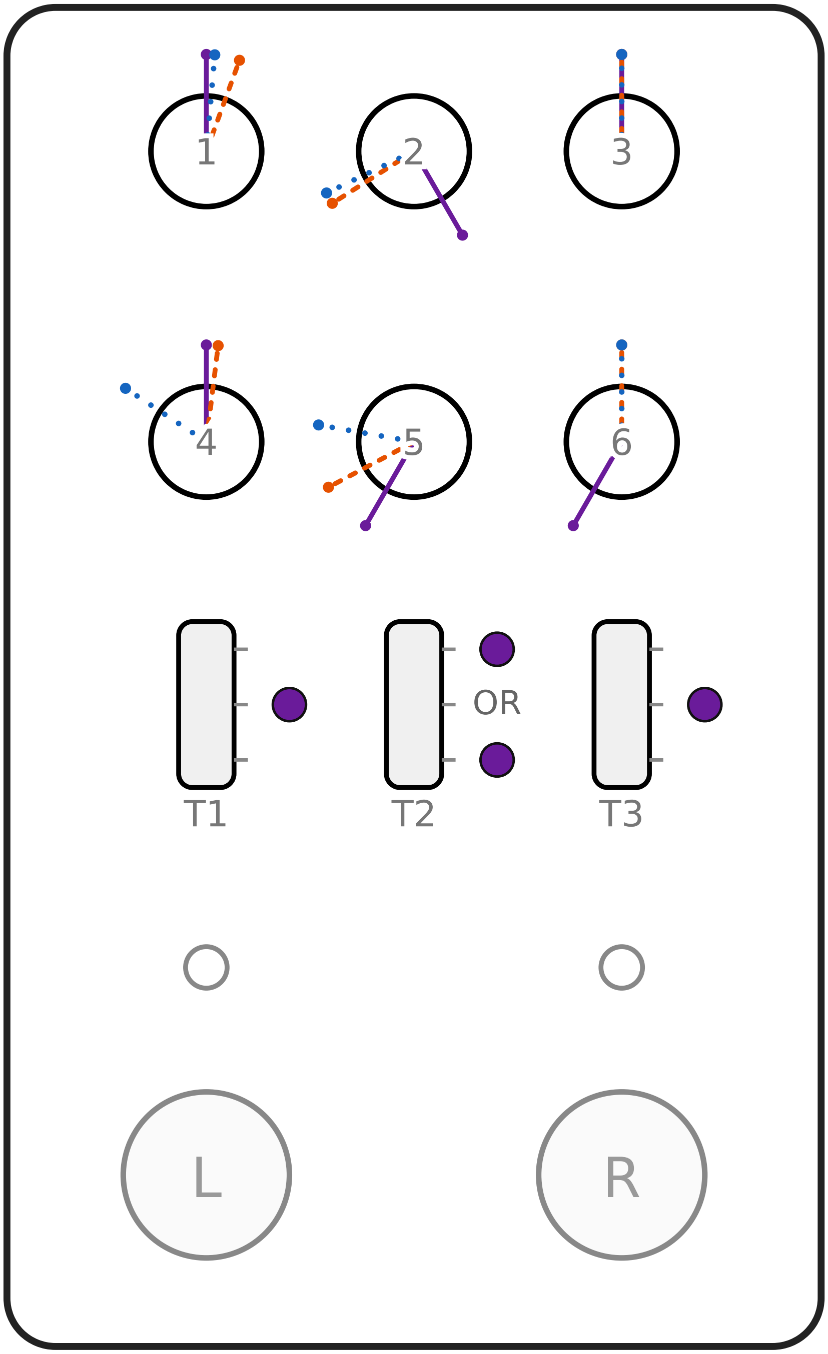
K1	F3	2900 Hz	47% · 11:40
K2	F3 bandwidth	62.5 Hz	12% · 8:09
K3	F3 amount	1.0	50% · 12:00
K4	Voices	5	56% · 12:37
K5	Detune	24 ct	40% · 11:00
K6	Grain	20 ms	50% · 12:00

Toggles

T1	Octave	0	Middle
T2	Memory page	Set 1 or Set 2	Up or Down
T3	Gate	medium	Middle

Fling *Light and quick*

No factory slot: save it to any page



Between Wukong and Rice on every formant, with a modest three-voice stack so the edge stays on the formants. The most usable of the eight under a band.

Fling: every position

Menu 1 (default)

K1	Vocal vol	unity	50% · 12:00
K2	Mix	full voice	100% · 5:00
K3	Master vol	unity	50% · 12:00
K4	Tone	flat	50% · 12:00
K5	Glide	0 ms	0% · 7:00
K6	Vocal size	as written	0% · 7:00

Menu 2 (hold RIGHT)

K1	F1	880 Hz	57% · 12:40
K2	F1 bandwidth	32.5 Hz	9% · 7:55
K3	F1 amount	1.0	50% · 12:00
K4	F2	1600 Hz	52% · 12:14
K5	F2 bandwidth	47.5 Hz	11% · 8:04
K6	F2 amount	1.0	50% · 12:00

Menu 3 (hold LEFT)

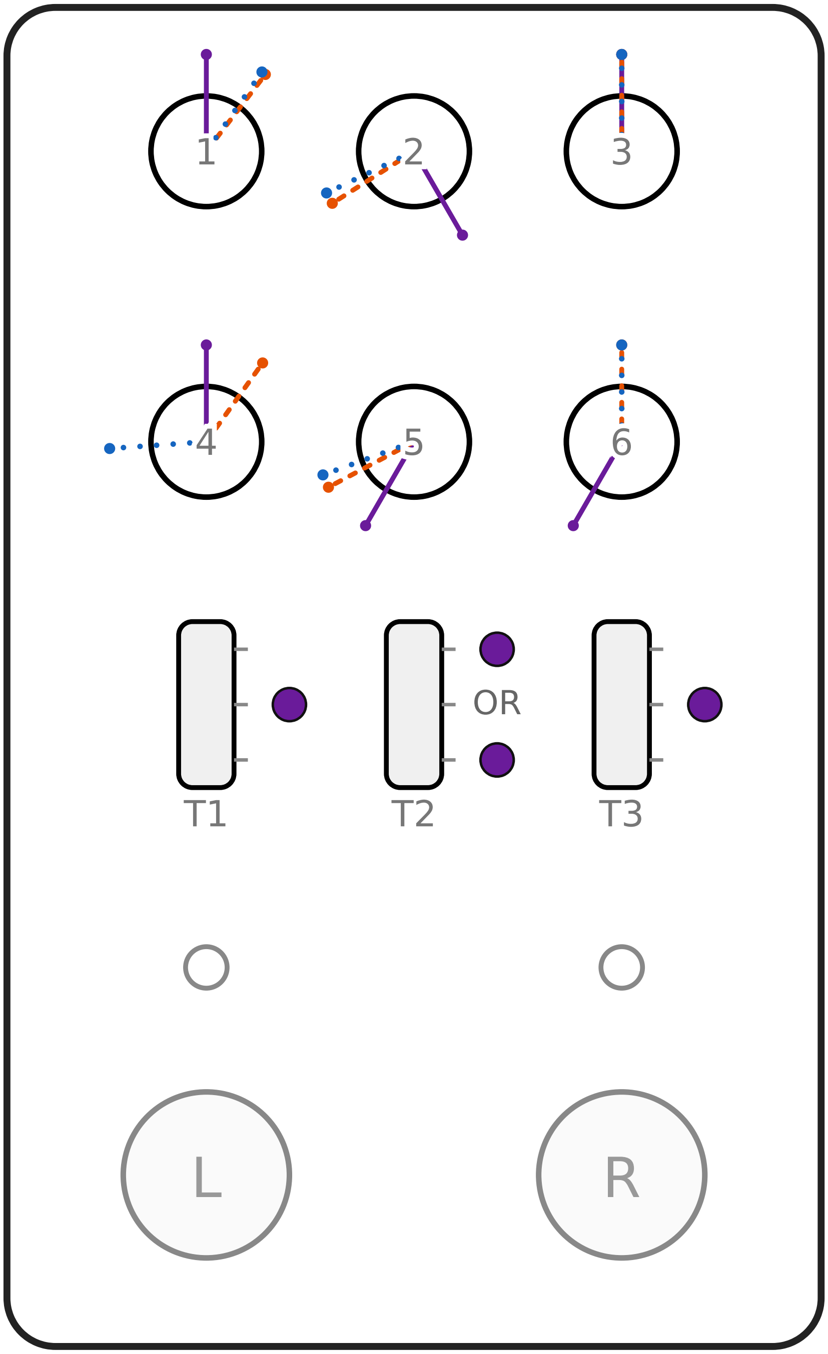
K1	F3	3050 Hz	52% · 12:10
K2	F3 bandwidth	62.5 Hz	12% · 8:09
K3	F3 amount	1.0	50% · 12:00
K4	Voices	3	31% · 10:07
K5	Detune	14 ct	23% · 9:20
K6	Grain	20 ms	50% · 12:00

Toggles

T1	Octave	0	Middle
T2	Memory page	Set 1 or Set 2	Up or Down
T3	Gate	medium	Middle

Ki-Ki *Small and furious*

No factory slot: save it to any page



The highest F2 of the eight and the thinnest stack of the eight, all three formants crowded high with nothing blunting them. Shrill, nasal, annoyed.

Ki-Ki: every position

Menu 1 (default)

K1	Vocal vol	unity	50% · 12:00
K2	Mix	full voice	100% · 5:00
K3	Master vol	unity	50% · 12:00
K4	Tone	flat	50% · 12:00
K5	Glide	0 ms	0% · 7:00
K6	Vocal size	as written	0% · 7:00

Menu 2 (hold RIGHT)

K1	F1	950 Hz	62% · 1:15
K2	F1 bandwidth	32.5 Hz	9% · 7:55
K3	F1 amount	1.0	50% · 12:00
K4	F2	1800 Hz	62% · 1:11
K5	F2 bandwidth	47.5 Hz	11% · 8:04
K6	F2 amount	1.0	50% · 12:00

Menu 3 (hold LEFT)

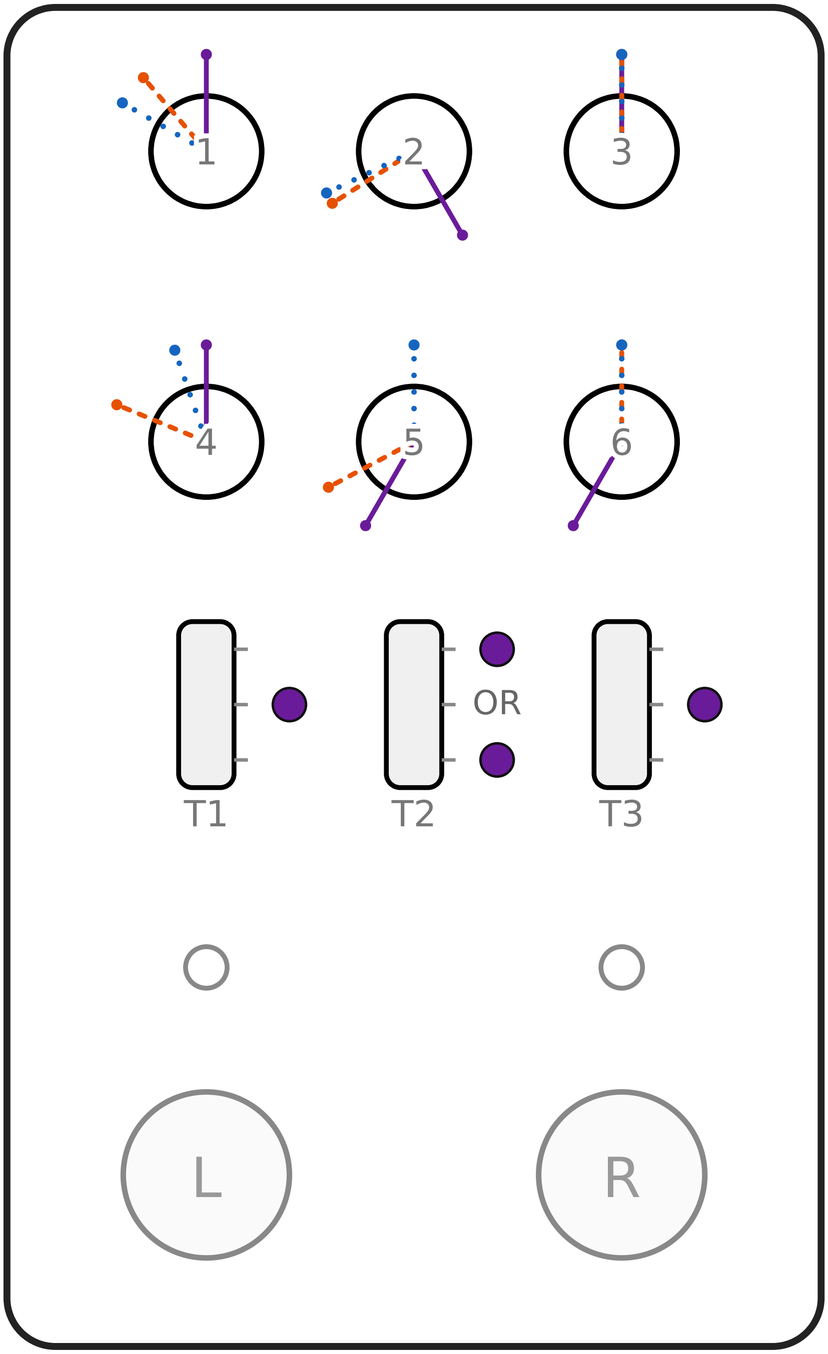
K1	F3	3350 Hz	62% · 1:10
K2	F3 bandwidth	62.5 Hz	12% · 8:09
K3	F3 amount	1.0	50% · 12:00
K4	Voices	2	19% · 8:52
K5	Detune	8 ct	13% · 8:20
K6	Grain	20 ms	50% · 12:00

Toggles

T1	Octave	0	Middle
T2	Memory page	Set 1 or Set 2	Up or Down
T3	Gate	medium	Middle

Master *Old and gravelled*

No factory slot: save it to any page



The lowest F1 and F3 of the eight, the widest detune of any card. Dark, frayed, faintly ridiculous.

Master: every position

Menu 1 (default)

K1	Vocal vol	unity	50% · 12:00
K2	Mix	full voice	100% · 5:00
K3	Master vol	unity	50% · 12:00
K4	Tone	flat	50% · 12:00
K5	Glide	0 ms	0% · 7:00
K6	Vocal size	as written	0% · 7:00

Menu 2 (hold RIGHT)

K1	F1	640 Hz	37% · 10:39
K2	F1 bandwidth	32.5 Hz	9% · 7:55
K3	F1 amount	1.0	50% · 12:00
K4	F2	1080 Hz	28% · 9:45
K5	F2 bandwidth	47.5 Hz	11% · 8:04
K6	F2 amount	1.0	50% · 12:00

Menu 3 (hold LEFT)

K1	F3	2400 Hz	30% · 10:00
K2	F3 bandwidth	62.5 Hz	12% · 8:09
K3	F3 amount	1.0	50% · 12:00
K4	Voices	4	44% · 11:22
K5	Detune	30 ct	50% · 12:00
K6	Grain	20 ms	50% · 12:00

Toggles

T1	Octave	0	Middle
T2	Memory page	Set 1 or Set 2	Up or Down
T3	Gate	medium	Middle

9 Every gesture

To do this	Do that
Engage or bypass	Tap either footswitch
Latch menu 2 (F1, F2)	Hold RIGHT ~1 s: it latches while you hold
Latch menu 3 (F3, voice stack)	Hold LEFT ~1 s: it latches while you hold
Jump to the other menu	With one latched, tap the other footswitch
Leave a menu	Tap the blinking side's own footswitch
Save to a slot	Hold one footswitch past 1 s, then press the other while still holding. The held side is the slot.
Charge	Engaged, no menu: stomp both together and hold
Firmware update	**Bypassed only:** press both together, hold ~2 s

TOGETHER, NOT STAGGERED

Both-footswitch gestures need both switches going down at the same time. Holding one for a second and then adding the other is the *save* gesture, and it will save instead.

What the LEDs mean

Single colour, both of them: the left LED is always blue, the right always orange. Nothing ever changes an LED's colour, only whether it is off, solid or blinking.

State	Left (blue)	Right (orange)
Bypassed	off	off
Left slot engaged	solid	off
Right slot engaged	off	solid
Freeform engaged	solid	solid
Menu 2 latched	off	blinking
Menu 3 latched	blinking	off
Saved	3 blinks	3 blinks
Save refused	one double-flicker, both	
Charging	alternating, speeding up	

10 When it misbehaves

No scream, just my guitar

Check an LED is solid. Then **knob 2 (Mix)**: at 7:00 you hear only dry signal. Then the **gate** (T3): on *high* with low-output pickups the voice may never open. Try the middle.

A knob does nothing

Working as designed. After every menu change and every recall the knobs are inert until moved. Nudge it further.

It chases the wrong note

Play one note at a time and mute what you are not using. Open strings ringing under a lead line are the usual culprit. Heavy drive in front makes tracking worse: put DBscreamZ earlier.

Both LEDs flickered, nothing saved

You were on Freeform. It has no slot. Flip T2 to Set 1 or Set 2 and save again.

I tried to save and got update mode

Or the reverse. The gestures differ by timing: save is hold one then add the other; update is both at once, from bypass.

It sounds thin and buzzy

Formant bandwidths set too wide will do that. The cards give the factory values: 7:55, 8:04 and 8:09.

The scream lags my playing

Check **glide** (knob 5). At 5:00 it takes 300 ms to reach each new note. At 7:00 it is instant.

11 Specifications

Platform	Cleveland Music Co. Hothouse, Daisy Seed3, 125B enclosure
Processing	48 kHz, 48-sample blocks, 480 MHz
Voice	FOF formant-grain synthesis, three formants, no noise sources
Tracking	Cycfi Q bitstream autocorrelation, monophonic
Bypass	Buffered, 10 ms crossfade on engage
Power	9 V DC, 2.1 mm barrel, centre negative
Memory	4 voice slots plus the global charge configuration, kept across power cycles

Firmware update

From **bypass**, press both footswitches together and hold about 2 seconds. The pedal appears over USB [universal serial bus] in DFU [Device Firmware Upgrade] mode. Flash it with the Daisy Web Programmer or dfu-util. Press the switches together, not one and then the other.

Credits

Formant voice synthesis derived from MonkSynth / Delay Lama. Pitch detection by cycfi/q, Boost Software License 1.0. Hardware from clevelandmusicco/HothouseExamples, open source hardware under CC BY-SA 4.0. The voices are synthesised from measured formant coefficients; no audio is redistributed.

Personal, non-commercial build. The names here are ours and refer to nothing.